Reported Operator Injuries and Wage Expenditure in U.S. Bus Agencies: A Fiscal-Year Panel, 2019–2024

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Abstract

Whether compensation responds to unsafe working conditions is an important question for transit workforce management, but annual administrative records do not directly measure contractual wages or employees' perceptions of reward. This study links National Transit Database labor, expense and monthly safety records for directly operated conventional bus service during 2019–2024. The screened panel contains 1,670 agency-years from 304 agencies. Exact consecutive fiscal-year matching yields a principal estimation sample of 1,345 observations from 284 agencies. Agency and year fixed-effects models relate prior reported operator injuries per 100,000 vehicle revenue hours to subsequent real wage expenditure per worked operating hour, adjusting for service scale, workforce composition and state labor-market wages. The originally planned all-staff specification estimates a coefficient of −0.000149 (95% confidence interval −0.008255 to 0.007957). An explicitly exploratory operator-specific measurement extension estimates −0.000917 (95% confidence interval −0.009713 to 0.007879). Neither supports the predicted positive annual association. All 20,040 screened agency-months pass employee-count identities and feasibility bounds comparing combined monthly counts with narrower major-event records and non-major worker summaries. Exact non-major occupational allocation is not independently observable. A major-event diagnostic likewise has an interval spanning zero. These estimates describe conditional within-agency associations with annual accounting expenditure, not identified wage-setting responses, causal hazard premiums, or tests of perceived effort–reward imbalance. Across the observed 10th-to-90th-percentile injury-rate span (0 to 1.540 injuries per 100,000 revenue hours), the exploratory operator coefficient corresponds to −0.14% (95% interval −1.49% to +1.22%) in expenditure per worked hour. The contribution is a reproducible accounting and fiscal-alignment framework that clarifies what the administrative records can and cannot reveal.

Keywords: transit workforce; operator compensation; reported injuries; National Transit Database; panel data

1. Introduction

Transit organizations must manage both workforce compensation and safety. These responsibilities are related but not interchangeable: spending more per labor hour may reflect contractual adjustments, overtime, worker composition or accounting allocation, while recorded injuries reflect only events meeting a reporting definition. A study connecting the two therefore needs to establish who is being paid, which injuries are counted and whether their reporting periods overlap correctly.

This paper asks: How is prior fiscal-year reported operator injury intensity associated with subsequent real wage expenditure per worked operating hour among U.S. agencies directly operating conventional buses? The question concerns annual adjustment within agencies. It is not a comparison of individual drivers' pay or an estimate of compensation for a worker's personal injury risk. Restricting the study to conventional bus service and direct operation preserves a common service and employer boundary; it deliberately excludes rail and purchased transportation.

Relevant scholarship examines risk–wage differences, transit employee compensation, and the effect of pay schemes on driving behavior. None of those literatures by itself identifies a wage adjustment caused by preceding agency injury reports. Section 2 differentiates these questions and the available evidence.

The question here runs in the other temporal direction: whether previously reported operator injury intensity co-varies with later real operator wage expenditure *within an agency* after accounting for common years and covariates. The candidate contribution is a reproducible U.S. multi-agency measurement and fiscal-alignment framework: aligning monthly injuries to actual employer fiscal years, separating operator from all-operating-staff wages, checking the scope of different injury reporting products, and preserving imprecise null estimates. This is **not** a novel compensation–safety topic, a test of a measured bargaining mechanism, or a causal identification strategy. Whether the narrower measurement-and-association contribution is sufficient for publication remains an external scholarly judgment.

2. Related literature and institutional evidence

2.1 Occupational risk, injury severity, and bargaining

The compensating-differential literature concerns wages paid for anticipated employment disamenities, not necessarily short-run wage changes following realized injuries. Hamermesh and Wolfe (1986, working-paper version; published 1990) distinguish the incidence of occupational wage loss from its duration. In their linked worker–industry data, injury-duration measures account for substantially more of the overall wage differential than event incidence, while interactions with union membership indicate greater compensation for injury incidence among union workers. Dorman and Hagstrom (1998) similarly emphasize the sensitivity of hazard-pay estimates to risk measures and labor-market institutions. Neither design observes the transmission of agency injury reports into subsequent collective bargaining settlements. The NTD exposure here counts reported operator injuries but lacks their duration, uncompensated earnings losses, and worker-specific perceptions of risk.

2.2 Transit pay arrangements and behavioral responses

Johnson, Reiley and Muñoz (2015) examine fixed versus passenger-based driver remuneration and bus operations in Santiago. Tiznado et al. (2014) survey 100 drivers at four consenting Santiago firms: 77% favor more bonuses linked to operational service, 59% oppose conditioning salary on a passenger-satisfaction index, and reported work arrangements include variable payments, paid allowances, and differences in shifts and working conditions. These are descriptive preferences in a small, non-national sample; their recounting of earlier accident comparisons is not a new causal estimate from that survey. Ramírez et al. (2022) randomize 157 students and transportation-sector professionals to payment treatments in a ten-stop computer experiment. Individual pay per passenger leads to greater dispersion of chosen speeds than fixed pay; team pay reduces opportunistic driving decisions while preserving substantially more regular speed profiles than individual payment. The simulated output difference between team and individual pay is not statistically significant. No real workplace injuries or collective bargaining settlements are observed.

Additional evidence concerns the reverse direction from compensation and working arrangements toward safety. Wu et al. (2019) report a cross-sectional retrospective survey of Chinese bus drivers in which demanding hours and highly performance-dependent pay are jointly associated with crash risk, but cannot be disentangled as separate causal exposures. The National Academies (2012) synthesize transit-agency reward and discipline practices intended to support safety. Together these studies warrant treating wage design, hours, and safety as potentially jointly determined. They do not provide an estimate of reported injuries leading to wage expenditure changes in U.S. bus agencies.

2.3 Organizational wage-cost studies using the NTD

Obeng and Ugboro (2025) provide an especially close administrative-data precedent. They estimate a theoretical joint wage, cost, and factor-share system on 108 selected single-mode U.S. bus systems in 2022, combining directly operated and partially contracted service in their sample. Their labor input price includes fringe benefits and differs from the operator-specific salary-expenditure ratio measured here. Employee effort and the proposed sequence from government performance incentives to employee wage premiums are not directly observed; the paper imposes a conceptual effort mechanism on system-level data. The analysis

does not include occupational injury counts or bargaining/ratification dates. Its existence rules out any claim that ours is the first NTD study of U.S. transit wages; the narrower candidate contribution is a multi-year fiscal-year-matched injury/expenditure association and a transparent demonstration of the data's occupational and timing limits.

2.4 Actual base rates, progression, bonuses, and bargaining dates

Primary employer records also show why annual wage expenditure is not a simple negotiated wage rate. CTA approved incentives in December 2022 consisting of $1,000 hiring bonuses, a 3% retention incentive on eligible hours through December 2023, and bus-operator starting-rate changes from roughly $24 to more than $28 per hour, with shortened progression toward top rates. Its announcement attributes these initiatives to staffing and recruitment pressures; this is not proof of an injury-induced premium (Chicago Transit Authority 2022). VTA announced on 6 January 2022 an ATU Local 265 agreement providing scheduled nominal increases of 2%, 4%, and 4% across three years and a one-time $3,500 payment. VTA separately approved a June 2022 amendment, effective for operator progression from 21 February 2022, shortening the progression period to 36 months (Santa Clara VTA 2022a, 2022b). Its 2022–2025 agreement additionally provides conditional accelerated progression for operators with no chargeable accidents and other eligibility criteria. These are source-documented examples of predetermined raises, workforce incentives, pay progression, and safety-conditioned rewards. They are not estimates of a risk-to-wage response.

2.5 Distinct question and identification limits

The present study examines whether previous fiscal-year reported operator injury intensity co-varies with the next year's accounting wage expenditure per worked hour within a directly operating bus agency. It neither estimates the equilibrium price of job risk nor the effect of paying incentives on actual injuries. The research can document the scope and precision of a national administrative association. It cannot by itself isolate the contract renegotiation channel, accident severity, operator turnover, hiring or retention responses, or the influence of wage rules on safety. Those mechanisms would require additional independent data and appropriate identification.

3. Conceptual framework and hypothesis

Compensating-differential theory motivates a positive relationship between undesirable job conditions and wages under particular labor-market conditions. Lavetti (2023) reviews why empirical estimates are difficult to interpret, including worker sorting, labor-market frictions and endogenous mobility. Employer fixed effects do not overcome every identification problem. Moreover, persistent differences in job amenities and annual responses to realized injury counts are different estimands.

The theory and the measured estimand must be distinguished. An equilibrium compensating differential concerns the wage paid for an expected job disamenity, given sorting and bargaining conditions. By contrast, the regressor here measures *realized reported injury events* in an earlier fiscal year. The dependent variable measures *aggregate wage expenditure divided by worked operating hours* in a later year. A positive within-agency annual association would require several links: reported events would need to change the perceived or negotiated value of risk, the agency would need discretion to modify pay within the observed horizon, and that decision would need to alter the accounting ratio rather than merely other dimensions of compensation. None of those links is measured directly.

The public transit labor context weakens any presumption of immediate adjustment. The National Academies (2023, ch. 4) documents scheduled wage increases, tenure-related scales and collective bargaining provisions that can determine the timing of pay changes independently of annual incidents. Baum et al. (2026) demonstrate, within Chicago's specific contractual environment, that paid hours and overtime opportunities can matter even without a change in base rates. More generally, Guardado and Ziebarth (2019) show that worker safety investments can complicate the sign of the risk–wage correlation. Hence the sign of our conditional coefficient is not uniquely predicted by the theory.

Contractual wage rigidity should be distinguished from union representation. A multi-year agreement can fix a wage schedule in advance while still requiring pay increases each year, and a later settlement may authorize retroactive payments. For instance, WMATA's agreement with ATU Local 689 specified annual general wage increases from July 2020 through July 2023. San Francisco's 2022–2024 operator agreement specified wage adjustments with contingencies tied to fiscal projections, as well as performance-related premiums. These are institutional examples, not evidence that injury reports triggered those provisions (WMATA 2019; SFMTA 2022).

Historical research also cautions against assuming that collective bargaining necessarily weakens compensation for risk. Hamermesh and Wolfe (1990) report stronger occupational injury-risk differentials in their union sample, whereas Dorman and Hagstrom (1998) report sensitivity to risk measurement and worker bargaining arrangements. Those studies concern a different wage–risk estimand and cannot identify a national transit bargaining effect. The present data do not contain verified bargaining-unit membership, dated negotiated base rates, contract renewal events or reports of employee risk perceptions.

The retained positive hypothesis was a plausible but restrictive operational expectation in the internally recorded original analysis plan, not a direct empirical implication of a fully specified structural model. Testing it remains useful for describing the particular observable one-year relationship, but its rejection or non-rejection cannot establish whether agencies compensate hazardous jobs.

The hypothesis recorded before the original regressions is retained without being rewritten to fit the results:

H1: Higher prior-year reported operator injury intensity is positively associated with subsequent real wage expenditure per operating labor hour within transit agencies.

The original operationalization of H1 covers all operating employees. The operator-specific outcome was developed after inspecting the original results and is treated as an exploratory measurement extension, not an independently confirmed hypothesis. Its purpose is occupational alignment, not the search for statistical significance.

Organizational-behavior research provides a related but distinct perspective. Rugulies and Krause (2008) examine perceived effort–reward imbalance and compensated musculoskeletal injuries in a longitudinal cohort of San Francisco transit operators. Their effort and reward measures are proxies; the reward items cover respect and job security, without monetary pay or promotion prospects. The study therefore provides psychosocial evidence rather than direct evidence about cash compensation. Our administrative expenditure ratio does not measure perceived fairness, recognition, security or effort–reward balance. Likewise, transit-event injuries are not a workers' compensation musculoskeletal outcome. We use this literature to specify a construct boundary, not to claim that H1 is a direct test of effort–reward imbalance.

Compensation can also precede safety outcomes, through incentives, selection or other organizational processes. The original plan included a reverse-direction exploratory association. Neither direction is causally identified here, and a one-year lag should not be interpreted as resolving simultaneity or omitted-variable bias.

4. Data and measures

4.1 Sources and screened cohort

Annual National Transit Database employee, agency-information and operating-expense records are linked to monthly modal safety and service data using agency identifiers, mode and service type. The unit is agency –fiscal year. The cohort is restricted to directly operated conventional bus service, coded MB/DO. It contains 1,670 screened agency-years from 304 agencies, including one Puerto Rico agency; 245 agencies are observed in all six years. This is neither all transit modes nor a complete census of all U.S. transit employers. Direct operation also does not establish government ownership.

The preceding audit requires usable wages and operating hours, reconciled accounting identities, absence of specified source quality flags, twelve unique matched safety months, positive service exposure and nonnegative injury counts. Missing observations are distinguished from observed zeros. The broad nominal expenditure-per-hour review band of $7.25–$150 is a diagnostic screen, not a legal compliance determination. Source coverage and screening outputs are retained in the reproducibility package.

Safety and labor records are aligned to the agency's actual fiscal-year end rather than automatically treating report years as calendar years. A lag requires the same agency, consecutive report years and adjacent twelve-month fiscal windows. Missing intervening years are not bridged. These rules yield 1,351 eligible exposure–outcome pairs. Removing six singleton-agency observations in fixed-effects estimation leaves 1,345 observations from 284 agencies, with outcome years 2020–2024.

4.2 Wage expenditure

The originally planned outcome is the log of real operating wage expenditure divided by all worked operating labor hours. The denominator sums vehicle operations, vehicle maintenance, facility maintenance and general administration, excluding capital labor. This measure includes operators and non-operators. Paid absences and fringe benefits are separately recorded; the main wage outcome does not include them.

The operator-specific extension divides operator wage expenditure by full-time plus part-time operator operating hours across the same four functions. Functional vehicle-operations hours are not used as a shortcut for occupational operator hours. Original 2019–2023 employee workbooks and the archived 2024 workbook provide the required occupational split. All 1,670 rows match; function-level identities reconcile and operator plus non-operator hours reproduce the original operating-hour totals exactly. No historical operator hours are imputed and no new agencies are added.

Both outcomes remain expenditure ratios, not contractual base wages or individual compensation. At an accounting level, total wage expenditure divided by total worked hours is an hours-weighted average of expenditure per hour across worker groups or pay categories; it can therefore change even when each contractual base wage is unchanged. Overtime premiums, workforce entry and exit, tenure mix, training, standby assignments, and the treatment of paid versus worked time can affect the ratio. None of these channels is separately identified by the annual aggregates. Nominal ratios are deflated using the average monthly national all-items CPI-U, not seasonally adjusted, over each fiscal window, expressed relative to the 2024 calendar-year average of 313.688833. This is a national purchasing-power adjustment, not a local cost-of-living measure.

NTD report year	Eligible agency-years	Median operator expenditure / worked hour	Median all-staff expenditure / worked hour
2019	282	$22.37	$23.79
2020	279	$22.87	$24.50
2021	282	$24.44	$25.80
2022	276	$25.57	$27.15
2023	275	$26.47	$28.03
2024	276	$27.88	$29.39

Note: Unweighted nominal cross-sectional medians. Agency composition varies over time. These are accounting expenditures per worked hour, not individual contractual wages.

4.3 Injury intensity and reporting audit

The principal exposure is previous fiscal-year monthly-reported operator injuries divided by vehicle revenue hours, multiplied by 100,000. Vehicle revenue hours measure service exposure, not operator labor exposure or an individual probability of injury. A separate extension uses operator labor hours as the denominator and therefore changes the coefficient's units.

The 2019–2024 safety manuals were reviewed for the non-rail injury criterion. The core criterion requires immediate medical attention away from the event scene; delayed treatment and unrelated illness are not a complete occupational injury census. We did not identify a change to this core bus criterion. That finding does not establish constancy of every event threshold, category, reporting practice or published aggregation. Expanded worker-assault reporting in 2023 includes events without injury and must not be equated with an increase in injury counts.

FTA distinguishes combined monthly modal data, individual major-event records and non-major summaries. The monthly modal product is not transformed to preserve historical reporting thresholds (Federal Transit Administration 2026). A crosswalk comparing the archived monthly numerator with current major-event operator injuries reproduces 1,441 of 1,670 agency-years exactly. Monthly totals sum to 7,379, whereas current major-event totals sum to 6,025; 229 agency-years differ. Non-major summaries contain worker injuries without a complete operator/non-operator split. Adding all non-major worker injuries produces discrepancies in 282 rows and is not a valid occupational reconstruction.

The products have different coverage, so equality is not a universal validation criterion. A fresh monthly extract reproduces all archived annual operator totals. Across 20,040 screened agency-months, total employee injuries equal operator plus other-employee injuries; combined operator and other-employee counts each equal or exceed their corresponding major-event counts; and the summed excess never exceeds non-major worker injuries. Every month passes these identities and feasibility bounds. The operator excess is 1,354, the other-employee excess 563, and 1,060 non-major worker injuries remain unallocated by these differences. These are implied components, not independently observed occupational assignments. Public non-major records do not allow an exact operator reconstruction. Vehicle location is not a valid substitute for occupation. The checks establish coverage consistency without proving complete occupational validity. No original injury values are changed; the narrower major-only diagnostic remains exploratory.

4.4 Covariates

Controls include previous-year log vehicle revenue hours; previous-year vehicle-maintenance, facility-maintenance and administration shares of operating labor hours, with vehicle operations omitted; previous-year part-time operating-hour share; and contemporaneous log real state private-sector average weekly wages. Full-time and part-time totals reconcile exactly. Quarterly Census of Employment and Wages records supply the state labor-market proxy. Twenty-seven fiscal windows ending in August or October use a within-quarter constant wage-rate approximation. CPI and market controls cover every screened observation. State wages are not occupation-specific or city-specific reservation wages.

5. Analytical strategy

The main model regresses log real wage expenditure per hour on prior injury intensity, organizational covariates, agency effects and report-year effects. Agency-years receive equal weight. The focal regression may be written as:

$$\log\left(W_{it} / H_{it}\right) = \alpha_i + \lambda_t + \beta R_{i,t-1} + \gamma' X_{i,t-1} + \delta M_{it} + \varepsilon_{it}$$

where W_it/H_it is real operator or all-operating-staff wage expenditure per worked hour, R_i,t-1 is the preceding contiguous fiscal year's reported operator injuries per 100,000 bus vehicle revenue hours, X_i,t-1 contains lagged service and workforce-composition controls, and M_it is the current fiscal-year state private-sector wage proxy. This is a conditional linear projection: beta is not an identified structural wage response. Agency effects remove stable differences; report-year effects absorb common annual shocks. The resulting slope describes conditional annual within-agency covariation, not permanent between-agency risk premiums.

Inference uses agency-clustered sandwich standard errors with a CR1 finite-sample adjustment and a t distribution with the number of agency clusters minus one degrees of freedom, following the methodological considerations discussed by Cameron and Miller (2015). These standard errors accommodate dependence

within an agency, but do not ensure independence across agencies or address endogeneity. Stronger geographic-shock adjustment is examined using state-by-year effects. A lagged dependent wage variable is not added to this short panel.

The internal original plan was recorded before regression estimation; it was not externally preregistered. Its principal hypothesis, ten sensitivity specifications and exploratory reverse-direction model are preserved. The operator extension plan was recorded after those results, but before fitting its models. All eight specified operator extensions and an identical-sample all-staff comparison are retained. A further major-event diagnostic was documented before fitting it, after examining the earlier models and reconciliation. These extensions are exploratory and are not independent replications.

The estimator residualizes fixed effects using rank-aware QR decomposition. An independent full-dummy SVD implementation checks the operator model coefficient and clustered covariance; maximum absolute differences are approximately 1.40×10^{-12} and 5.47×10^{-14}. Sample membership, ranks, residual degrees of freedom, all coefficients and source hashes are archived.

6. Results

6.1 Measurement and descriptive results

Every screened row has verified operator hours and positive operator wage expenditure. Annual median nominal operator expenditure per worked hour rises from $22.37 in 2019 to $27.88 in 2024. These are unweighted cross-sectional medians in a changing panel, not a balanced-worker wage-growth estimate. The all-staff median is higher in each annual cross-section. The occupational distinction therefore changes the level of the measure even though it does not change the main inference.

In the archived combined series, 58.0% of screened agency-years have zero reported operator injuries. Ninety-four agencies report zero throughout their observed records, and 205 agencies have within-agency injury-intensity variation in the screened 2019–2024 panel. In the exact O1 estimation sample, 777 of 1,345 lagged rates (57.77%) are zero, and 196 of 284 agencies have more than one observed prior-year injury-rate value. The predictor has mean 0.525, standard deviation 1.065, median zero, 75th percentile 0.669, and 90th percentile 1.540 reported operator injuries per 100,000 vehicle revenue hours. Zeros are retained; they indicate no injuries recorded under the monthly series definition in that period, not an absence of occupational harm.

6.2 Original and operator-specific estimates

The originally planned all-staff model estimates −0.000149 (clustered standard error 0.004118; 95% confidence interval −0.008255 to 0.007957; p=.971). Re-estimating that outcome on the operator extension's identical sample reproduces the original coefficient. The predicted positive association is not supported.

The operator-specific model estimates −0.000917 (clustered standard error 0.004469; 95% confidence interval −0.009713 to 0.007879; p=.838). A one-unit increase in prior intensity—one reported injury per 100,000 vehicle revenue hours—is associated with approximately −0.092% in subsequent real operator wage expenditure per worked hour, with an exponentiated 95% interval of approximately −0.967% to +0.791%. This interval includes both increases and decreases; it does not establish equivalence to zero. Over the actual estimation sample's 10th-to-90th-percentile span of 1.540 injuries per 100,000 vehicle revenue hours, the corresponding fitted difference is 100[exp(beta x 1.540)-1] = −0.14%, with the coefficient-based 95% interval transformed to approximately **−1.49% to +1.22%**. This is a linear-model contrast at a realistic change in the regressor, not a causal change in pay. The interval does not bound multi-year or delayed responses.

Model	Specification	N	Coefficient	95% CI	p
O1	Operator outcome; service-hour	1,345	-0.00092	[-0.00971, +0.00788]	0.838

Model	Specification	N	Coefficient	95% CI	p
	injury rate				
C1	All-staff outcome; same observations	1,345	-0.00015	[-0.00826, +0.00796]	0.971
O2	Operator outcome; operator-hour injury rate	1,345	-0.00076	[-0.01129, +0.00978]	0.888
O3	Balanced cohort	1,225	+0.00067	[-0.00743, +0.00877]	0.872
O4	Neither year in 2020 or 2021	516	+0.01170	[-0.00777, +0.03117]	0.238
O5	Fiscal windows end before 2023	807	+0.00121	[-0.01197, +0.01438]	0.857
O6	State by year effects	1,345	+0.00335	[-0.00574, +0.01243]	0.469
O7	Log one plus injury rate	1,345	-0.00830	[-0.02944, +0.01283]	0.440
O8	Exclude operator wage changes over 50 percent	1,336	+0.00115	[-0.00710, +0.00940]	0.784
M1	Major-event-only operator injuries; exploratory	1,345	+0.00182	[−0.00757, +0.01121]	0.704

Note: Agency- and report-year fixed effects with agency-clustered CR1 uncertainty, unless otherwise specified. O1–O8 and M1 are exploratory checks; C1 is the original all-staff model re-estimated on the same sample. O2 and O7 change predictor units; coefficients should not be directly compared.

6.3 Sensitivity and reporting-product diagnostic

All originally planned sensitivity intervals include zero. The operator extensions examine the labor-hour injury denominator, balanced cohort, removal of pairs involving 2020 or 2021, fiscal windows ending before 2023, state-by-year effects, log-transformed injury intensity and exclusion of annual operator wage changes exceeding 50% in absolute value. Every focal interval spans zero. These are related specifications using overlapping data, not independent confirmations. Transformed exposures and alternative denominators should not be compared as if their coefficients shared a common unit.

The major-only diagnostic uses the operator-injury field in 33,989 current MB/DO event records during 2018–2024. Incident keys are unique and relevant date and operator-injury fields are present. Serious-injury fields are not added to total injuries. Major-event absence is treated as zero only within fiscal windows with complete archived monthly safety reporting. The model estimates 0.001817 (standard error 0.004771; 95% interval −0.007574 to 0.011208; p=.704), on 1,345 observations from 284 agencies. It addresses a narrower injury construct and is not a replacement confirmatory estimate.

The original reverse-direction exploratory coefficient is 0.133744 (standard error 0.220326; p=.544), relating prior log real all-staff wage expenditure to log(1 + injury intensity). It is not an injury-count elasticity and does not demonstrate a protective or harmful compensation effect.

7. Discussion

The analysis does not detect the hypothesized positive annual within-agency association. Improving wage measurement from all staff to operators preserves that conclusion, as does the narrower major-event diagnostic. The monthly identities and coverage bounds establish internal consistency, but neither those checks nor statistical agreement independently validates the non-major occupational allocation. Inference remains limited to the specified administrative associations and publisher-defined injury categories.

The proposed contract-timing mechanism is further complicated by the construction of the outcome. Research using administrative payroll information distinguishes workers' base wage rates from overtime premiums and bonuses, which may move differently (Grigsby, Hurst, and Yildirmaz 2021). The annual NTD ratio can increase through overtime premiums or a change in the mix of operators even when negotiated base rates do not change. Conversely, safety concerns can lead to operator-protection clauses or safety-linked premiums without a general wage increase. In a 2022 San Francisco operator agreement, eligibility for certain performance premiums included a record without preventable collisions, while assault-related insurance benefits were also addressed (SFMTA 2022). These provisions are not a direct response to the NTD operator-injury count used here.

An alternative contemporaneous pathway arises if staffing pressure and longer shifts are associated with both injury risk and realized labor spending. Dembe et al. (2005) found elevated injury hazards among workers on overtime schedules in a U.S. longitudinal sample, whereas Matre et al. (2021) reported imprecise pooled associations for overtime defined categorically, with stronger evidence for very long working hours and low overall certainty. These findings motivate a separate same-fiscal-year descriptive question but do not resolve its direction of causation in public transit. No contemporaneous extension has been estimated as part of the present manuscript.

The organizational interpretation has several boundaries. Stable differences in persistent risk and compensation are absorbed by agency effects. The one-year lag is an administrative matching choice, not a claim that bargaining, worker perceptions, risk information, or contractual rates adjust within twelve months. The National Academies (2023) documents wage progression and negotiated compensation arrangements; Baum et al. (2026) illustrates how duty selection, paid-hour guarantees and overtime affect realized operator earnings in one transit system. None of those mechanisms is observed or tested in this agency-level panel. Annual reporting may miss delayed contract adjustment. Realized injury counts may not represent anticipated risk. The data do not measure bargaining dates, union agreements, worker tenure, overtime composition, workforce demographics or perceived reward. Contract rigidity and worker sorting are possible explanations for weak annual associations, but the present results do not establish them. A null estimate should not be redescribed as support for an unmeasured mechanism. At the model's observed exposure scale, the operator confidence interval narrows plausible one-year linear associations relative to an unbounded claim, but its width still admits modest positive and negative associations. Non-detection is not statistical equivalence, and the 2019–2024 record is too short to interpret as a long-run hazard premium.

The policy implication is correspondingly modest. Managers should not use aggregate wage expenditure as evidence that workers are adequately compensated for risk, nor use this study to conclude that compensation cannot influence safety. Wage-setting, incentive design and injury prevention are different processes. More informative organizational research would connect dated compensation agreements or exogenous wage changes to occupationally specific injury records, and would measure worker perceptions if effort–reward balance is the intended construct.

The design is vulnerable to shared shocks affecting staffing, service, expenditure and reporting. Lagging the exposure does not eliminate these pathways. The pandemic is especially important in a six-year panel; exclusions reduce observations and cannot reproduce conditions in a longer pre-pandemic record. Source quality screens and the requirement for monthly reporting restrict generalizability. National CPI and state private-sector wages do not fully represent local operator labor markets. Cross-agency residual dependence

may remain despite year and state-year controls. No formal equivalence, causal identification or long-run adjustment analysis is claimed.

A practical implication is methodological: annual wage-expenditure statistics should not be used as a proxy for a transit agency's responsiveness to incidents without checking fiscal dates, operator accounting, and contractual arrangements. The study contributes a fiscal-alignment and occupational accounting framework with a bounded annual-association result. It does not establish the organizational mechanisms connecting safety information to pay-setting. Connecting dated compensation agreements, staffing responses or worker-level perceptions to occupational injury records would permit more informative tests of those mechanisms.

A source-matched illustration underscores the accounting limitation. CTA's reported operator wage expenditure per worked hour rose from approximately $35.84 in its 2022 fiscal year to $38.11 in 2023, then declined to $35.95 in 2024; VTA's corresponding nominal ratio rose from $40.65 in its fiscal year ending June 2022 to $43.32 in 2023 and $44.31 in 2024. These ratios are derived directly from the archived NTD operator wage expense and worked-hour inputs, are not base-rate schedules, and are not estimates of the causes of wage changes. CTA's documented 2022–2023 bonuses and progression changes and VTA's scheduled raises make clear that expenditure movements can coexist with numerous pay-setting mechanisms. Neither agency was selected on the sign or magnitude of its regression residual; these are institutional illustrations selected because signed or official wage documentation could be verified.

8. Conclusion

A fiscal-year-matched 2019–2024 panel provides no clear annual relationship between prior reported operator injury intensity and subsequent real wage expenditure per operating hour in directly operated U.S. conventional bus services. The operator outcome reconciles to labor accounts and the combined safety numerator satisfies monthly coverage bounds, although exact non-major occupational allocation remains unobservable in the public summaries. The results do not identify a hazard premium, establish adequate compensation or directly test effort–reward imbalance. The accounting framework and transparent null results support further research; stronger claims require better measures of pay-setting and organizational mechanisms and fuller documentation of occupational injury allocation.

Reproducibility and AI-assisted workflow

A fresh 14 October 2026 rerun of the archived preparation and model scripts from the original raw inputs reproduced the exact 1,670-row, 140-variable operator panel, all nine operator-model rows, model-membership keys, and reported focal coefficients and CR1 uncertainty. The original all-staff and operator-specific fixed-effects estimates are unchanged. Internal consistency and executable reproducibility do not establish independent validation of FTA classification decisions, agreement mechanisms, or causal identification. No new contract-response regressions are reported in this revision.

The accompanying research package retains source extracts and acquisition records, internal analysis plans, measurement audits, model specifications, exact model membership, computational checks and tables. Archived original results are not overwritten. Public agency-level aggregate records are used; there are no individual employee identifiers in the analytical panel. A public repository link and any applicable institutional ethics determination must be finalized by the authors before submission.

ChatGPT/Codex assisted with source retrieval, literature discovery, code generation, numerical analysis and manuscript drafting. The package preserves the generated code and checks. This disclosure does not represent human scholarly verification as already complete. Human authors must independently review the sources, methods, analyses and text and take responsibility for the submitted version. Final authorship, funding, conflicts and contribution statements require author confirmation and are not fabricated here.

The user-accessible interlibrary-loan copies of Obeng and Ugboro (2025) and Tiznado et al. (2014) were used for scholarly review. These are not redistributed in any public reproducibility archive. Their copyright

and access conditions must be respected. The research reproducibility bundle is separate from the user's private literature library.

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Appendix A. Analysis history and variable scope

The original plan, dated September 17, 2026, preceded the original regression estimates but followed the source audit. The operator plan, dated October 1, 2026, followed original results and preceded operator estimates. The major-event check plan followed the operator results and safety reconciliation and preceded its own estimation. None is a public preregistration. The plans should be evaluated with that chronology, not described as independent confirmatory tests.

Agency-year wage expenditure ratios are not employee-level wages. Service-hour injury intensity is not a personal injury probability. The operator-labor-hour variant changes exposure units. Major-event counts omit injuries outside that product’s scope. Agency and year fixed effects address stable and common observed-period variation, not every time-varying confounder. All focal model estimates, including unsupported hypotheses, are retained.

Appendix B. Post-review contract-timing feasibility protocol

Motivation and chronology

This protocol follows the original 2019–2024 analysis, the post-original operator-specific analysis, and the 11 October 2026 academic feedback. It is neither a revision of the originally planned hypothesis nor a preregistered confirmatory study. The protocol is a feasibility exercise; no extension coefficients are reported here.

Candidate data structure

A new dataset would identify the employing agency and precise bus-operator bargaining unit, each operative collective bargaining agreement, wage adjustment and step-progression dates, contract ratification and effective dates, employer-approved amendments, COLA rules, bonuses and retroactive payments, safety-related premiums, and safety protections. Only verified occupation-matched agreements would be coded. Agreements for purchased-transportation contractors must not be assigned to the agency’s directly operated employees without a documented employer crosswalk.

Data feasibility and temporal coverage

The archive supports descriptively eligible samples of 1,659 same-year, 1,345 one-year-lagged, 1,060 two-year-lagged and 775 three-year-lagged agency-year observations after excluding singleton agencies. These

are eligibility counts, not regression findings. The two- and three-year designs would have only four and three outcome years, respectively, with high prevalence of recorded zero injuries. Any lagged models prompted by the review must be presented as exploratory and evaluated as a complete prelisted set rather than selected for statistical significance.

Proposed substantive analysis

A bargaining-episode sample would compare operator base rates or contracted step schedules following a settlement with risk information available before negotiation. An initial 15–25 agency pilot would assess document recoverability, signed status, occupational alignment, overlap with NTD windows and the number of independent bargaining episodes. Underlying contract timing is not assumed exogenous. Without sufficient events, agreement coverage or credible operator wage measures, the extension should be reported as a descriptive contract study rather than a causal model.

Interpretation limits

The new design could investigate whether some safety-related concerns enter renegotiation or instead lead to nonwage provisions, but injury-event reports are incomplete proxies for perceived risk. A union indicator alone is inadequate for measuring wage rigidity or bargaining timing. Researchers should avoid extrapolating a causal hazard premium from fixed-effects interactions and should disclose any contract coding uncertainty and nonrandom availability of public agreements.